

Wei Lu

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EDUCATION

Rotman School of Management, University of Toronto , Toronto, Canada <i>Ph.D. in Quantitative Marketing</i>	Expected 2023
Toulouse School of Economics , Toulouse, France <i>M.Sc. in Economics (Economic Theory and Econometrics, Doctoral track)</i>	2017
Central University of Finance and Economics , Beijing, China Victoria University , Melbourne, Australia <i>B.A. in Economics and B.Bus., joint degree (with distinction)</i>	2015
University of California, Los Angeles , Los Angeles, California, USA Exchange student	2014

RESEARCH INTEREST

Topic: Economics of Platforms and Algorithms

Method: Structural Modeling, Causal Inference, Machine Learning, Natural Language Processing

WORKING PAPERS

Product Development and Platform Design (Job Market Paper)

(First author; with Avi Goldfarb and Nitin Mehta)

First-place Winner, PDMA Doctoral Dissertation Proposal Competition, 2022

Winner, ISMS Doctoral Dissertation Proposal Competition, 2021

Peer Effects from Friends and Strangers: Evidence from Random Matchmaking in an Online Game

(with Daniel Goetz) [\[SSRN\]](#)

Accepted at EC'22. Recipient of TD MDAL Research Grant, 2021

Under review at Management Science

Increasing Dynamic App Engagement with Peak-End Effects (with Daniel Goetz)

Recipient of BEAR/BI-Org Ph.D. Research Awards, 2021

Evaluation Inflation: Consumers Give Higher Product Ratings on Transaction Sites than Non-transaction Sites (with Claire Tsai and Ying Zeng)

Recipient of TD MDAL Research Grant, 2021

Revise and Resubmit at Journal of Marketing Research

Motivated Reasoning in the Field: Polarization in Precedent, Prose, and Policy in U.S. Circuit Courts

(First author; with Daniel L. Chen)

Media Coverage: Washington Post

WORK IN PROGRESS

Food Purchasing Patterns and Nutrition Quality in the Time of Pandemic: Evidence from COVID-19 and Lockdowns

(Co-first author with Guanlan Hu; with David Soberman and Mary L'Abbé)

CONFERENCE PRESENTATIONS

The 23rd ACM Conference on Economics and Computation (EC'22)*, Boulder, CO	2022
ISMS Marketing Science Conference, Chicago, IL	2022
TD MDAL Research Roundtable, Toronto, ON, Canada	2022
ISMS Marketing Science Conference, Rochester, NY	2021
Trans-Atlantic Doctoral Conference, London, UK	2021
EMAC Conference, Madrid, Spain* (* presented by coauthor)	2021
The 2018 Meetings of The Canadian Law and Economics Association, Toronto, ON, Canada	2018

GRANTS, HONORS & AWARDS

First-place Winner, PDMA Doctoral Dissertation Proposal Competition	2022
Doctoral Completion Award	2022
Winner, ISMS Doctoral Dissertation Proposal Competition	2021
BEAR/BI-Org Ph.D. Research Awards (CAD \$6900)	2021
TD MDAL Research Grant (CAD \$5000, with Daniel Goetz)	2021
TD MDAL Research Grant (CAD \$5000, with Claire Tsai, and Ying Zeng)	2021
NBER Digitization Tutorial Fellow	2021
ISMS Marketing Science Conference Doctoral Consortium Fellow	2020
NBER Young Scholars Workshop on the Economics of Artificial Intelligence Fellow	2019, 2020
Quantitative Marketing and Structural Econometrics Workshop Participant	2019
University of Toronto Fellowships	2017-2023
Director's Scholarship	2017-2019
Best Undergraduate Thesis Award	2015
Outstanding Graduate of Beijing City	2015

TEACHING EXPERIENCE

Project Coach, University of Toronto

Management Analytics Practicum (<i>Master of Management Analytics</i>), Instructor: Dmitry Krass	2021
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Teaching Assistant, University of Toronto

Pricing (<i>Undergraduate</i>), Instructor: Matthew Osborne	2020
Statistics for Management (<i>MBA</i>), Instructor: Sridhar Moorthy	2020, 2022
Introduction to Big Data Analysis (<i>MMA</i>), Instructor: Daniel Goetz	2019-2021
Analytics for Marketing Strategy (<i>MMA</i>), Instructor: Nitin Mehta	2019
Principles of Marketing (<i>Undergraduate</i>), Instructor: Zachary Zhong	2019
Consumer Behaviour (<i>Undergraduate</i>), Instructor: Scott Hawkins	2019

MISCELLANEOUS

Computer Skills: Python, R, MATLAB, Stata, SPSS, SAS.

Languages: English (fluent), Mandarin (native), French (beginner)

REFERENCES

Avi Goldfarb (Supervisor)

Rotman Chair in Artificial Intelligence and Healthcare

Professor of Marketing

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ABSTRACTS

Product Development and Platform Design (Job Market Paper)

Wei Lu, Avi Goldfarb, and Nitin Mehta

Recipient of ISMS Doctoral Dissertation Award 2021 and PDMA Best Doctoral Dissertation Proposal Award 2022

Abstract: Digital mobile app and video game platforms often adopt a 30% commission rate on all consumer purchases. Such a high rate has been challenged both in public and in court, as one key argument is that a 30% rate hurts the innovation incentives of software developers. Nevertheless, empirical evidence of how commission rates affect innovations remains scarce. Using a unique dataset of video games in the Early Access Program on Steam, we study how commission schemes can change both the product update decisions and the release decision to Steam’s main Store of developers. We first use transformer-based neural network language models and supervised machine learning algorithms to classify updates into Feature, Regular, and Bug fix updates. Then, we document positive correlations between past product updates and consumer demand and between past demand and updates and current release decisions. Motivated by these facts, We construct and estimate a structural model of developer and consumer behaviors on platforms and find substantial costs for producing updates and releasing the product. We further examine the impact of varying platform commission rates on product development decisions and find that while lowering commissions can lead to more updates, fewer games will be released, leading to an over-investment in Early Access.

Peer Effects from Friends and Strangers: Evidence from Random Matchmaking in an Online Game (with Daniel Goetz)

Extended Abstract at EC’22. Recipient of TD MDAL Research Grant, 2021

Under review at Management Science

Abstract: We empirically study how to seed products in a network in the presence of peer effects from both friends and strangers. The product adoption decisions of friends and strangers can both strongly influence consumers’ purchase behaviour, but their importance relative to each other is unknown. We first collect novel data on microtransactions in an online gaming app to quantify the lift in purchase probability from observing friends’ and strangers’ product adoption choices. To isolate exogenous exposure to peers’ product adoptions, we leverage the app’s quasi-random matchmaking system. Observing a friend’s product adoption has twice the marginal effect on purchase probability as observing a stranger’s product adoption, but the cumulative effect from observing strangers’ product adoptions is much larger over time. We then calibrate a diffusion model using the in-app social network of 9.4 million players. We find that seeding products to active players who interact with more strangers generates almost four times the number of net new adoptions compared to seeding to players who are more central in the social network.

Increasing Dynamic App Engagement with Peak-End Effects (with Daniel Goetz)

Recipient of BEAR/BI-Org Ph.D. Research Awards, 2021

Abstract: The peak-end effect in consumer psychology predicts that individuals will re-engage with an activity sooner if the activity ends with a relatively positive experience. Using millions of unique usage sessions from two online apps, we document that an exogenously allocated positive experience makes consumers more likely to end a usage session with the app, even as it shortens the duration until they begin their next usage session. Our results show that leveraging the peak-end effect leads to shorter but more frequent usage sessions, creating a tradeoff in-app engagement.

Evaluation Inflation: Consumers Give Higher Product Ratings on Transaction Sites than Non-transaction Sites (with Claire Tsai and Ying Zeng)

Recipient of TD MDAL Research Grant, 2021

Revise and Resubmit at Journal of Marketing Research

Abstract: Prior research in online reviews has focused on describing their general distributions (e.g., J-shape). The present research complements this literature by comparing rating distributions on transaction sites versus non-transaction sites and identifying conformity to perceived norms as the underlying mechanism. In six experiments with randomization ($N = 1,977$) and archival data analysis across eight platforms ($N = 275,851$ products based on approximately 500 million reviews), we observe an evaluation inflation effect: average ratings and share of perfect scores (5-star reviews) are higher on transaction sites (e.g., Amazon, DoorDash) than non-transaction sites (e.g., Goodreads, Yelp) or when no platform is mentioned. This is because consumers infer the norm of online reviews on transaction sites from sellers' desire to receive perfect scores and their impression of the substantial share of perfect scores. They conform to the norm by increasing their ratings. Causal evidence lends support to this conformity mechanism. Our experiments control for and rule out factors suggested in prior literature that may influence rating distributions (e.g., selection biases and frauds). This research contributes to the emerging consumer psychology literature in review creation and conformity in a digitized marketplace.